

1.2 Classwork: Speed, Velocity, and Unit Conversion

Reference (provided). average speed = $\frac{\text{distance}}{\text{time}}$

$$1 \text{ mi} = 5280 \text{ ft} \quad 1 \text{ mi} \approx 1.61 \text{ km} \quad 1 \text{ m} \approx 3.28 \text{ ft} \quad 1 \text{ ft} = 12 \text{ in}$$

$$1 \text{ km} = 1000 \text{ m} \quad 1 \text{ in} = 2.54 \text{ cm} \quad 1 \text{ h} = 3600 \text{ s} \quad 1 \text{ min} = 60 \text{ s}$$

1. A student walks 24.0 meters down the hallway in 18.5 seconds.
 - (a) Find the average speed in meters per second, rounded to three significant figures.
 - (b) How many significant figures does your answer have, and why not more?
2. Convert each speed. Write each conversion as a fraction and show the units cancelling.
 - (a) A bus on Via del Corso travels at 30.0 kilometers per hour. What is this speed in meters per second?
 - (b) A train travels at 25.0 meters per second. What is this speed in kilometers per hour?
3. A fastball leaves the pitcher's hand at 95 miles per hour.
 - (a) What is this speed in feet per second?
 - (b) What is this speed in meters per second?

4. Three vehicles: a car at 60 miles per hour, a train at 95 kilometers per hour, and a motorcycle at 27 meters per second. Convert all three to meters per second, then list them from slowest to fastest.

5. The speed of sound in air is 343 meters per second. You see a lightning flash and hear the thunder 4.0 seconds later. How far away was the lightning, in meters? In miles?

6. Our building is a New York brownstone, and the classroom is on the 6th floor — so from the entrance you climb 5 floors. Each floor is one flight of 18 steps, and each step is 8.0 inches high.
 - (a) How many steps do you climb in total?

 - (b) What is the total elevation gain, in feet? In meters?

 - (c) Climbing steadily at 1.2 steps per second, how long does the climb take? Give your answer in seconds, and again in minutes and seconds.

 - (d) What is your average vertical speed, in feet per minute? In meters per second?

7. Challenge: a car's speedometer reads 30 miles per hour. A student converts this to kilometers per hour and writes 48.28032 km/h. Explain what is wrong with this answer, and write a better one.